Note from Flint:

No matter if you are delivering oral presentation or writing a manuscript, make sure you understand your audience and your objective of the conversation (yes, oral presentation and manuscript writing are all considered conversation). You want to make sure your audience gets the right key message as accurately as possible. It is **your job** to help ease the understanding of your audience. After all, if your audience did not get what you wanted to deliver, then it is a waste of everyone's time, **including yours**. So, try to make life easier for your audience (in particular, your reviewers). Always imagine if you were the audience who is listening/reading your conversation for the first time, did you get the necessary information/background/context to capture the key message?

A. Presentation Template (to a first-timer audience)

A1. Narrative Arc

1. **Problem context** → what? why now? who cares?
2. **Objective** → precise research question & scope
3. **Constraints & Challenges** → why it's hard (if it is not hard, then that explains why nobody is doing it)
4. **Solution (Key Idea)** → intuition first, then mechanism (why do you design your solution this way? Killing a mosquito with a nuclear bomb is a solution; but is it a good solution?)
5. **Evidence** → theory (do you like to argue the solution is guaranteed to work, even before any experiments) & experiments (do you want to say from experiments: it just works)
6. **Relevance** → takeaways, limitations, future work

A2. Slide outline

1. Motivation (pain point/use case)
2. Problem definition (inputs, outputs, setting)
3. Key limitation in prior work (better to give a table)
4. Your key idea (better to use diagram)
5. Method (or a flow chart; get ready to explain design choices)
6. Theory highlight (necessary assumptions; theorem statement; interpretation; intuitions)
7. Experiment design (objective, benchmarks, baselines, metrics)
8. Results (do you get what you want to get? do experimental results support claims in the narrative?)

9. Takeaways (what are the implications of your results? who and how to use your research?)

B. Manuscript Template (conference)

Target: ICML/NeurIPS/ICLR/AAMAS style. Use this outline strictly unless you know what you are writing.

1. **Abstract (5-sentence formula)**
   S1 *Problem*: what and why now.
   S2 *Gap*: why prior work is insufficient.
   S3 *Key idea*: the mechanism in one sentence.
   S4 *Results*: theory findings (if any) + headline empirical gains with numbers (if any).
   S5 *Impact*: broader implication (some papers ignore this… I guess it depends on S4 and the length of the abstract)
2. **Introduction**
   - Context → Gap → Our idea → Contributions (bulleted, 3–5 items, each verifiable).
   - (optional but highly beneficial) End with a *Figure 1 overview diagram*.
3. **Related Work (Positioning)**
   - Organize by concepts, not a literature dump.
   - Comparative table with *Strengths vs. Limitations* for 3–5 key papers.
   - You want to smartly **choose** the strengths and limitations to help with your own narrative.
   - See attached Table 1 for an example.

2 Related Work

Method	AH	MLU	ISM	SEI	MF
PAVG[AISTATS'22]	✗	✔	✔	✔	✔
FedSARSA[ICLR'24]	✗	✔	✔	✔	✔
FRD[GLOBECOM'22]	✗	✗	✔	✗	✗
SCCD[TKDD'23]	✗	✔	✔	✔	✔
FedHQL[AAMAS'23]	✔	✗	✗	✔	✔
DPA-FedRL[TNNLS'24]	✔	✗	✗	✗	✔
FedHPD (ours)	✔	✔	✔	✔	✔

   - Table 1: **Comparison** of **FedRL methods**. AH denotes agent heterogeneity; MLU denotes multiple local updates; ISM denotes independence from server MDPs; SEI denotes sample efficiency improvement and MF denotes model-free.
   - Final paragraph: "We differ from **X/Y** by **[specific mechanism/theory]**."
4. **Problem Setup (I always have this section in my paper, and I like to call it "Problem Formulation")**

    - Notation, setting, assumptions.
    - Formal objective(s) and evaluation metrics.
    - Technical challenges can be discussed in detail here.
    - Threat/shift models if relevant (privacy, robustness, non-stationarity).
5. **Method**
    - Intuition → Algorithm → Complexity → Implementation details (ready to reproduce).
    - Pseudocode (dont make the pseudocode look too scary).
    - Design choices justified against challenges (callbacks to the technical challenges you have identified).
6. **Theory (optional but preferred)**
    - Assumptions (state clearly; explain necessity; some assumptions can be powerful tools to make analysis easy, but they are the weakest points to attack by "too strong assumptions"; so you have to prepare justification in advance to **preempt reviewers' attack**).
    - Lemmas → Theorems → Corollaries (each with 1–2 lines of interpretation).
    - Proof sketches in main; full proofs in Appendix.
7. **Experiments**
    - **Hypotheses** (H1–H3): state your design objective first.
    - **Benchmarks,** baselines (related work?), **Metrics** (primary/secondary), **Compute budget**.
    - **Protocols**: seeds, hyper-param ranges, early stopping, hardware.
    - **Main results** (tables/curves), **Ablations**, **Sensitivity/Robustness**, **Generalization** tests.
    - Report **effect sizes**, **CI**, **statistical tests**; release **seeded scripts**.
8. **Discussion**
    - Why it works, failure modes, limitations, ethical considerations.
9. **Conclusion**

- Re-state contribution, actionable future work.

Appendix

- Full proofs, additional experiments, dataset details, hyper-params, negative results.

B1. Fill-in Language Snippets

- **Problem statement.** "We study **[task]** under **[setting]**. The goal is to **[objective]** subject to **[constraints]**."
- **Gap framing.** "Prior work achieves **[strength]**, but struggles when **[condition]**, due to **[cause]**."

- **Key idea.** “Our insight is to **[mechanism]**, which addresses **[challenge]** by **[effect]**.”
- **Theory claim.** “Under **[assumptions]**, we prove **[guarantee]** (Thm X). Practically, this implies **[takeaway]**.”
- **Result headline.** “On **[benchmarks]**, **[method]** improves **[metric]** by **[Δ%]** over the strongest baseline, with **[p-value/confidence]**.”
- **…**
